$$f(\alpha_i) = \beta_i$$

$$P(x) = \prod_{i=1}^{n} (x - \alpha_i)$$

$$P(\alpha_i) = 0$$

$$P_i(x) = \frac{P(x)}{x - \alpha_i} \qquad P_i(\alpha_j) = 0$$

$$W_i(x) = \frac{P_i(x)}{P_i(\alpha_i)} \cdot \beta_i$$

$$W(x) = \sum_{i=1}^{n} W_i(x)$$

$$W_i(\alpha_i) = \beta_i$$

$$W_i(\alpha_j) = 0$$